

Temperature trends in Geneva

Our progress, findings and next steps

CME project group / 15 September 2026

We are investigating whether a simple linear trend adequately describes Geneva's annual mean temperature record, and how residual dependence, changing variance and possible structural change affect inference. This is our ongoing working record, based on the current files in our assignment folder.

Our current result. We estimate an upward trend of **0.0337°C per decade**. We have produced the residual diagnostics and calculated **DW = 1.4147**. The Monte Carlo test and Breusch–Pagan test are still unfinished.

1. Our data and current scope

We use Station 08: Geneva, Switzerland. The supplied metadata identifies NOAA NCEI's GHCN Monthly Temperature Version 4, QCF adjusted series, as the source. Each annual mean requires at least ten valid monthly observations [1].

Recorded observations	226
Recorded period	1753–2013
Missing calendar years	35
Incomplete supplied rows	0
Duplicate years	0
Mean temperature	9.5034°C
Minimum (1851)	7.6483°C
Maximum (2011)	11.7300°C

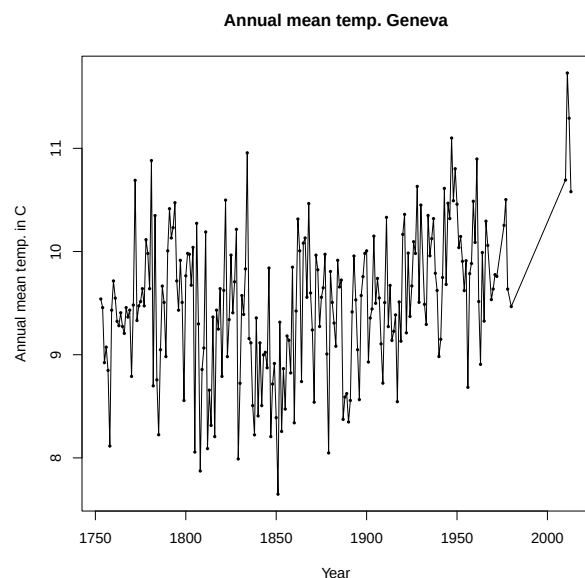


Figure 1: Our original temperature-series plot. The line connects available observations, including across gaps.

Our main data caveat: all years from **1981 to 2009** are absent. Other missing years are 1931, 1968, 1973–1975 and 1979. Our complete-case filter removes no rows. The connecting line from 1980 to 2010 does not represent observed annual movements in the missing period.

What we have saved: data preparation, OLS estimates, six plots and observed DW. Our script ends immediately after the DW calculation. We have not yet implemented formal Monte Carlo calibration, BP, a structural-break search, bootstrap inference or the Part III experiment.

2. Our linear trend model

We model annual mean temperature as a deterministic straight-line mean plus an error:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad x_i = \frac{\text{Year}_i - 1900}{10}.$$

We use i to distinguish the observation's position from its calendar year. Our transformation makes 1900 equal to zero and a ten-year increase equal to one unit. We transform time, not temperature. Thus β_0 is the fitted mean temperature in 1900 and β_1 is the trend in degrees Celsius per decade.

How we estimate it

We use `lm(Temperature ~ t, data = dat)`. Ordinary least squares chooses the intercept and slope that minimize

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^{226} (y_i - \beta_0 - \beta_1 x_i)^2.$$

Our fitted equation is

$$\hat{y}_i = 9.61663 + 0.0337093x_i.$$

Quantity	Estimate
Intercept	9.6166°C
Slope per decade	0.03371°C
Slope per century	0.3371°C
R^2	0.1054
Adjusted R^2	0.1014
Residual standard error	0.6596°C
Residual degrees of freedom	224

We read R^2 as the share of variation around the sample mean explained by the fitted line: approximately 10.5%. Our residual standard error measures the estimated scale of deviations around that line.

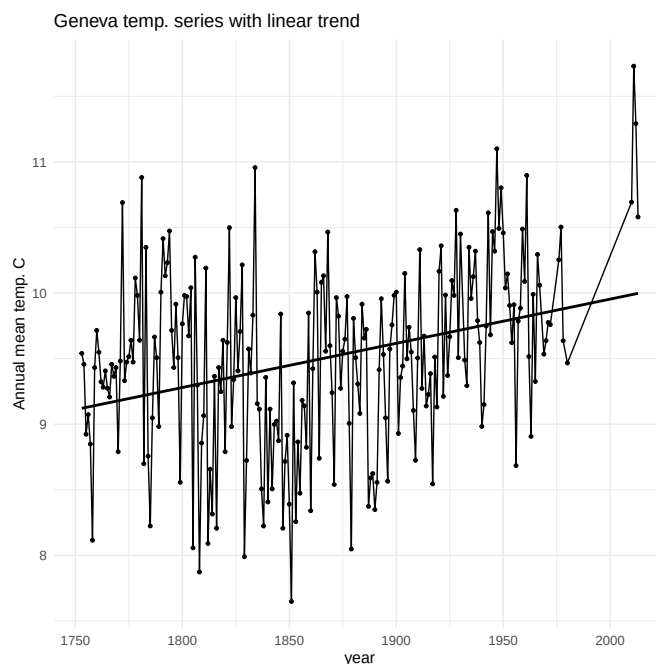


Figure 2: Our temperatures and fitted linear trend. The fit uses all 226 supplied observations.

What we can say at this stage

We find a positive average slope over the supplied record. The fitted line rises by about **0.8764°C** across 1753–2013, a 260-year span. This is a model-implied change, not the observed endpoint difference. Substantial variation remains around the line, and the four observations in 2010–2013 are all above it. Our printed OLS summary includes a conventional slope p -value of about 6.09×10^{-7} . We do not use it as a final significance conclusion: the residual diagnostics question the usual inference assumptions, and Part I explicitly postpones trend-significance conclusions [2].

3. Residual patterns and dependence

We define a residual as observed minus fitted temperature:

$$\hat{\varepsilon}_i = y_i - \hat{y}_i.$$

A positive residual means a warmer observation than our model predicts. A negative residual means a cooler observation. We use the residual time plot to look for remaining patterns and the ACF to examine dependence across observations.

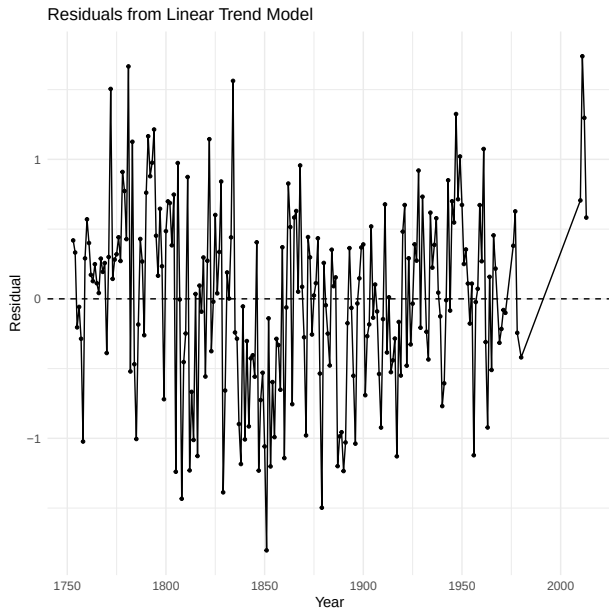


Figure 3: Our residuals over calendar time, with a zero reference line. Units are °C.

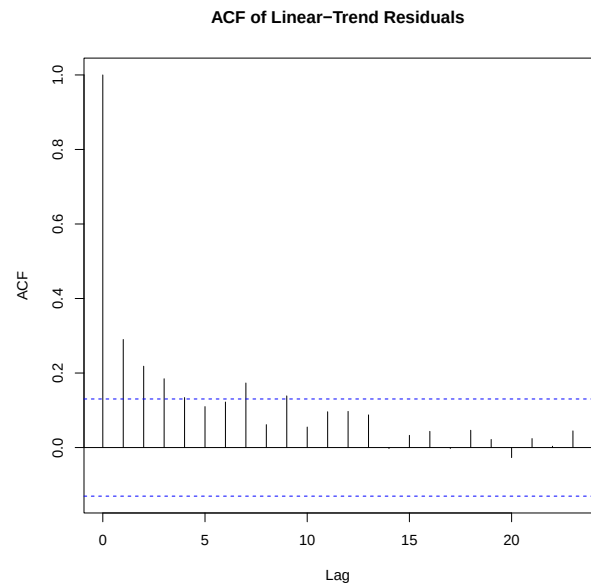


Figure 4: Our residual ACF. Lags refer to adjacent available observations.

What we see in these plots

We see runs of residuals on the same side of zero, including a negative stretch around the 1830s–1850s and positive residuals at the sample end. We interpret these patterns as reasons to investigate serial dependence and the adequacy of the mean trend. The time plot alone cannot distinguish those explanations or establish a structural break.

The ACF measures the association between residuals and lagged residuals. The values underlying our plot are approximately **0.290 at lag 1**, **0.218 at lag 2** and **0.185 at lag 3**. They exceed its approximate upper reference band of 0.130. We regard the bands as pointwise graphical guides. Lag zero equals one by construction.

The Durbin–Watson statistic we have calculated

Our function at the end of `main.R` computes

$$d = \frac{\sum_{i=2}^n (\hat{\varepsilon}_i - \hat{\varepsilon}_{i-1})^2}{\sum_{i=1}^n \hat{\varepsilon}_i^2}, \quad d_{\text{obs}} = 1.4147.$$

If adjacent residuals tend to move together, the numerator becomes smaller. The rough relation $d \approx 2(1 - \hat{\rho}_1)$ explains the agreement with our positive lag-one ACF. Values below two suggest positive dependence; values above two suggest negative dependence. These are interpretive guides, not critical values.

Our present conclusion is graphical and descriptive. We have not calculated a Monte Carlo rejection region or p -value. Also, both `acf()` and `diff()` use observation order: they treat 1980 and 2010 as consecutive observations. We cannot consistently call this lag “one calendar year.”

4. Residual magnitude and distribution

We use two further plots to examine changing residual magnitudes and the shape of the residual distribution. Neither is a formal test in our current script.

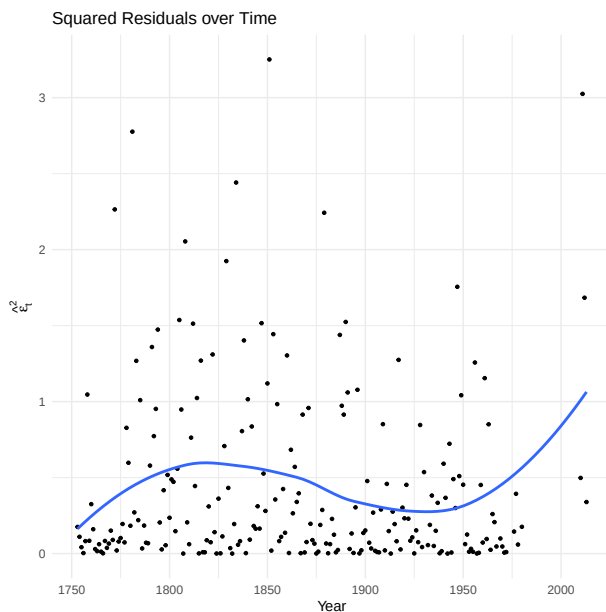


Figure 5: Our squared residuals with a LOESS smoother. Squared residuals have units of $(^{\circ}\text{C})^2$.

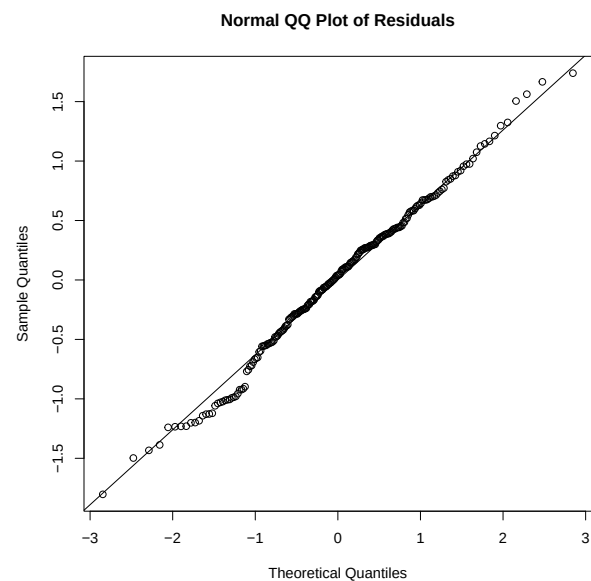


Figure 6: Our normal Q–Q plot, comparing ordered residuals with normal quantiles.

Squared residuals and LOESS

We square residuals to examine the size of deviations without their signs. LOESS fits local regressions to show a smooth pattern in those magnitudes. Our curve rises into the early nineteenth century, declines toward the early twentieth century, and increases near the sample end.

We see possible variation in residual magnitude, but we have not established heteroskedasticity. A wrong mean specification can also generate large squared residuals. The recent increase requires particular caution because there are only four observations after 1980, separated from the earlier data by a long gap. The smooth curve across that gap does not supply missing evidence.

Normal Q–Q plot

In a Q–Q plot, we compare ordered residuals with theoretical normal-distribution quantiles. An approximately straight pattern is consistent with an approximately normal distribution. Most of our points follow the reference line reasonably closely, with some departures in the tails.

We do not treat this appearance as proof of normality. It also tells us nothing conclusive about independence or constant variance. Normality is not needed simply to compute the OLS coefficients.

Our combined assessment so far

- We have an upward fitted trend, but the line leaves substantial variation unexplained.
- Our strongest current diagnostic concern is positive residual dependence, reflected in the ACF and observed DW.
- We see possible changes in residual magnitudes, but our BP test remains to be done.
- We need to distinguish dependence in the errors from remaining structure in the mean, particularly before choosing a bootstrap procedure.
- We have no estimated break date and no statistical evidence from a completed break test yet.

5. The formal tests we still need to finish

Durbin–Watson Monte Carlo calibration

We have implemented the observed statistic only. The handout requires a two-sided Monte Carlo test with $B = 9,999$ and $\alpha = 0.05$ [2]. Our next implementation needs to:

1. State the null error model, including the distribution, independence and variance assumptions. For example, an iid Gaussian null is stronger than merely assuming no first-order correlation.
2. Keep our observed regression design fixed, simulate errors under that null, and generate a simulated sample.
3. Refit the trend in every replication and calculate DW from the fitted residuals. Using raw simulated errors would omit the effect of estimating the trend.
4. Obtain separate 2.5% and 97.5% critical values, c_L and c_U . We will reject when $d_{\text{obs}} < c_L$ or $d_{\text{obs}} > c_U$, without assuming symmetry around two.
5. Report both critical values, the Monte Carlo p -value, our tail-count convention and the conclusion. Our script already sets seed 20260914, but no simulation currently uses it.

For example, we can document an equal-tail rank convention using

$$p_L = \frac{1 + \#\{d_b \leq d_{\text{obs}}\}}{B + 1}, \quad p_U = \frac{1 + \#\{d_b \geq d_{\text{obs}}\}}{B + 1}, \quad p_{\text{MC}} = \min\{1, 2 \min(p_L, p_U)\}.$$

These formulas describe planned work; we have no simulated numerical results yet.

The Breusch–Pagan auxiliary regression

We still need to test for a systematic linear relationship between residual variance and our time regressor:

$$\hat{\varepsilon}_i^2 = \gamma_0 + \gamma_1 x_i + v_i, \quad H_0 : \gamma_1 = 0 \quad \text{versus} \quad H_1 : \gamma_1 \neq 0.$$

The handout specifies $BP = nR_{\text{aux}}^2$, using the R^2 from this auxiliary regression, not from the temperature regression. We will compare it with an asymptotic χ_1^2 distribution and report its upper-tail p -value. We have one nonconstant variance regressor, which explains the one degree of freedom.

A failure to reject would not exclude every nonlinear variance pattern. We also need to qualify the usual chi-squared calibration in light of our residual dependence. The LOESS figure is not a completed BP test.

A point we need to clarify in Question 6

The required DW procedure simulates a finite-sample null distribution, whereas the specified BP calculation uses a large-sample approximation. We should not claim BP is inherently impossible to simulate. With fixed regressors and a specified iid Gaussian error model, both statistics are invariant to the regression coefficients and a common error scale. Monte Carlo calibration of heteroskedasticity tests is possible under suitable distributional assumptions [3]. We need to clarify the intended distinction with the lecturer; homoskedasticity alone does not specify a full finite-sample distribution.

6. Our next steps and working record

Our immediate priorities

- ☐ Resolve the calendar-time versus observation-index convention and document the missing years in figures and dependence measures.
- ☐ Complete the DW Monte Carlo test and BP auxiliary regression, record their assumptions and numerical outputs, and interpret them together with our plots.
- ☐ Write our overall assessment of the initial model, keeping estimates, graphical indications and formal testing conclusions distinct.

What we have not started in Parts II and III

- ☐ **Structural change:** search a continuous broken-trend model over admissible break dates with 15% trimming; report the RSS profile, best-fitting calendar break and observed RSS reduction.
- ☐ **Bootstrap inference:** justify a primary procedure and a sensitivity procedure using the diagnostics; use at least 999 replications for the reported break tests; construct percentile and percentile- t slope intervals while re-estimating the break.
- ☐ **Simulation study:** include no-break and break scenarios and vary at least two additional scenario dimensions; compare at least three bootstrap procedures with at least 1,000 Monte Carlo replications per configuration and 499 bootstrap replications within each, unless constraints are documented.

We maintain the fuller, editable checklist in `TODO.md`.

Reproducibility and provenance

We verified the unchanged `main.R` in a clean R 4.6.1 session with `ggplot2`. All six rendered plots reproduced the saved `Rplots.pdf` exactly. We captured the run in `docs/current-output.txt`. The data summaries and ACF values in this note were checked directly against the same data and fitted model; we have not represented them as separately implemented tests.

We run `make analysis` from the repository root and rebuild this document with `make report`. Our LaTeX source embeds the original plot PDF pages directly. The report text is a dated snapshot, so we must refresh its numbers and interpretation when we change the analysis. Default `source()` may skip display of bare `ggplot` expressions; our verified route is `Rscript -vanilla main.R`.

Sources

- [1] Supplied `Station08.csv` and `Station08_metadata.pdf`. NOAA NCEI, GHCN Monthly Temperature Version 4. DOI: [10.7289/V5XW4GTH](https://doi.org/10.7289/V5XW4GTH).
- [2] Marina Friedrich, *Computational Methods in Econometrics: Investigating Temperature Trends, 2026–2027*. Supplied `CMEAssignment2026Handout.pdf`.
- [3] Dufour, Khalaf, Bernard and Genest (2004), *Simulation-based finite-sample tests for heteroskedasticity and ARCH effects*, Journal of Econometrics 122, 317–347. [Research paper](#).